



Rice New Energy Fund Analytics Project

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Team Lead

The Team & Expertise

Rice New Energy Fund (RNEF)

~200k AUM, Sustainable Revolution

Rice New Energy Fund (RNEF) is a student-managed investment fund that manages a portfolio of public stocks related to the energy transition. Their 6 securities span 3 sectors: transportation, power, and industry. RNEF currently uses deep analysis and due diligence along with help from an algorithm to determine the appropriate amount of shares to buy, but they have not incorporated portfolio optimization analysis yet.

- Maximize returns subject to stochastic constraints
- Outperform the iShares Global Clean Energy Index
- Create a Full-Stack app that RNEF analysts can use as a tool to aid in the fundamental analysis process



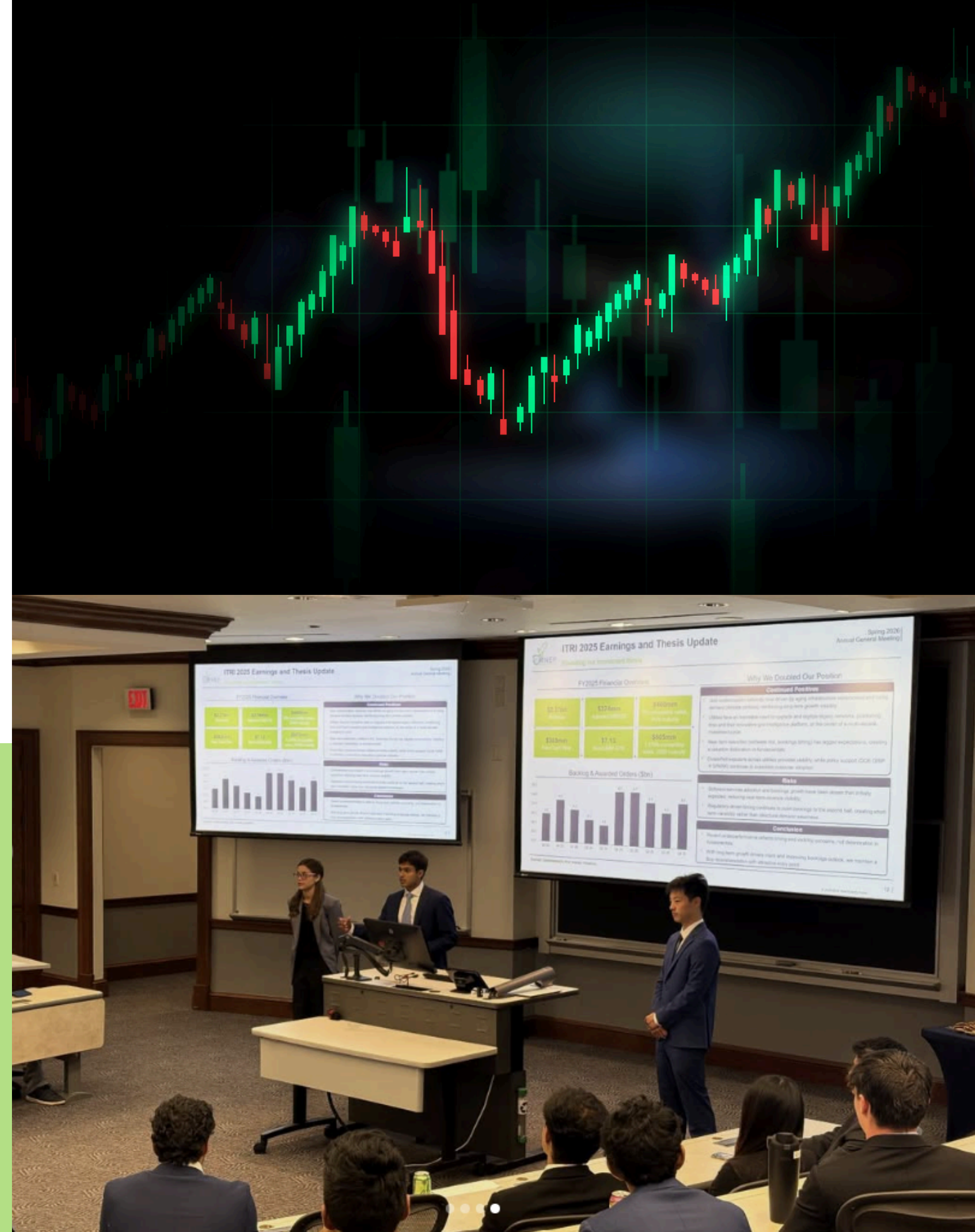
RNEF Portfolio Value



Motivation

RNEF currently...

- Uses analyst research and values stocks based on fundamental analysis
- No portfolio management
- Sets a list of objectives/targets that each stock has to hit before they sell



User Dashboard

 **RNEF Analytics**
Rice Renewable Energy F...

BACKTESTING

- Dashboard
- New Backtest**
- Strategy Library

New Backtest

VaR-Scaled

Equal Weight

TICKER UNIVERSE (10 active)

Deselect all

NEE

ENPH

FSLR

SEDG

BEP

PLUG

RUN

CSIQ

CEG

AES

Add extra ticker...

+ Add

Click grey tickers to toggle - Blue = added for this run - Min 3 required

START DATE

10/13/2022

END DATE

04/12/2026

REBALANCING FREQUENCY

Monthly

TRANSACTION COST

0 bps

10 bps

50 bps

MAX POSITION WEIGHT

10%

40%

60%

Walk-Forward Simulation

Prevents look-ahead bias

☒

Naive 1/N benchmark — equal allocation across all assets.

Min Volatility

Global minimum variance — focuses on reducing portfolio volatility.

Data Sources

Price data via yfinance

Risk-free rate via FRED TB3MS

Holdings from Excel tracker

Price data fetched live via Yahoo Finance on each backtest run

Walk-forward simulation rebalances at each period start using only prior data. 10 bps transaction costs applied.

Optimization Strategies

Set parameters and select strategies to simulate

BACKTEST NAME

e.g. Walk-Forward Q1 2026

STRATEGY

☒ Max Sharpe (MVO)

☐ Min Volatility

☐ HRP

☐ VaR-Scaled

☐ Equal Weight

TICKER UNIVERSE (10 active)

Deselect all

NEE ENPH FSLR SEDG BEP PLUG RUN CSIQ CEG AES

Add extra ticker...

+ Add

Strategy Guide

● RNEF Actual

● Max Sharpe (MVO)

Mean-variance optimization with Ledoit-Wolf covariance shrinkage.

● HRP

Hierarchical Risk Parity — no matrix inversion, robust to estimation error.

● VaR-Scaled

Position sizes inversely proportional to 95% historical VaR.

● Equal Weight

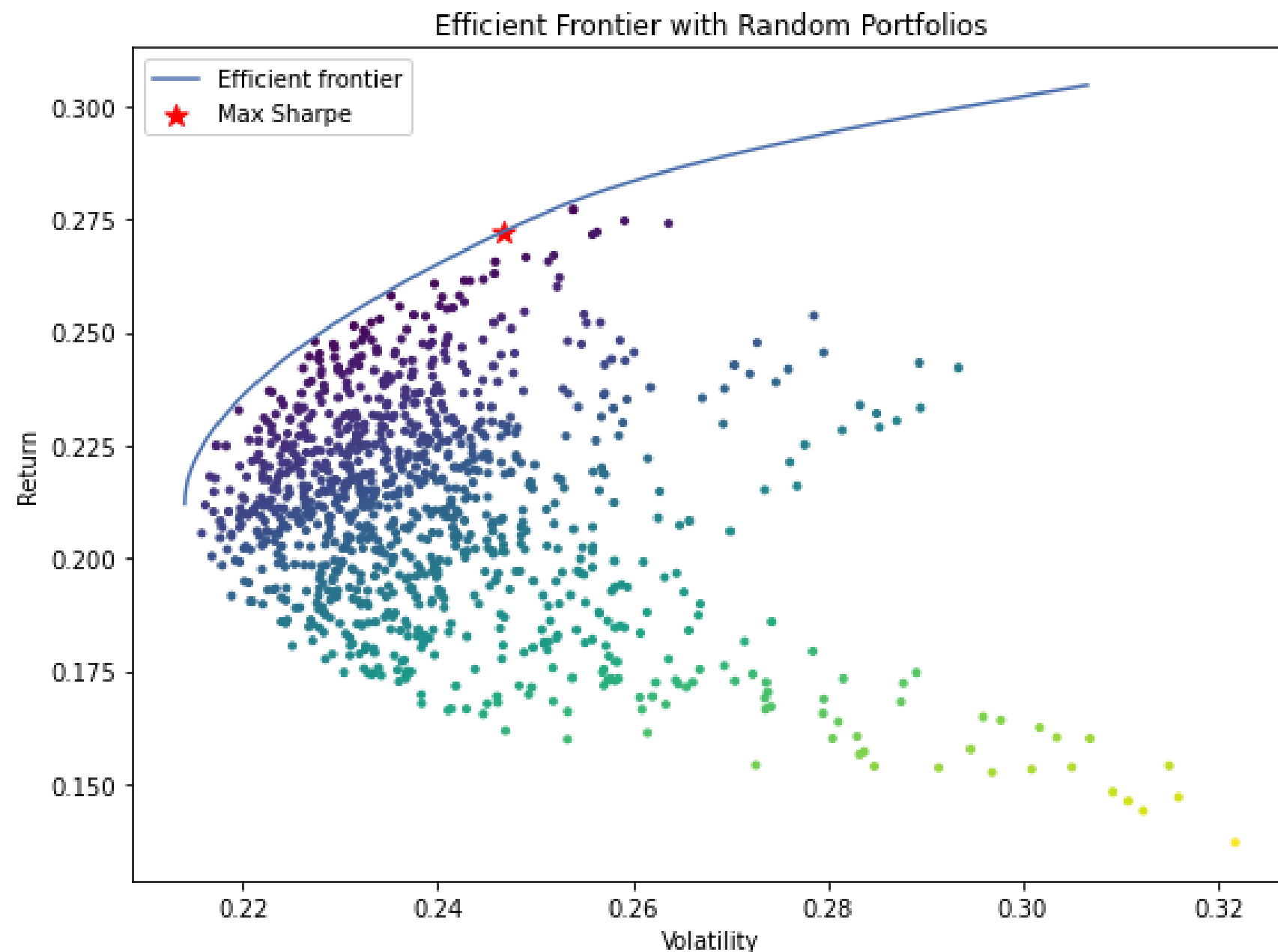
Naive 1/N benchmark — equal allocation across all assets.

● Min Volatility

Global minimum variance — focuses on reducing portfolio volatility.

Max Sharpe (MVO)

Optimizes for the highest expected return per unit of risk; traditional mean-variance approach that identifies the asset weights that maximize reward relative to historical volatility, constructing the mathematically optimal "tangency portfolio."



$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

R_p = Return of portfolio

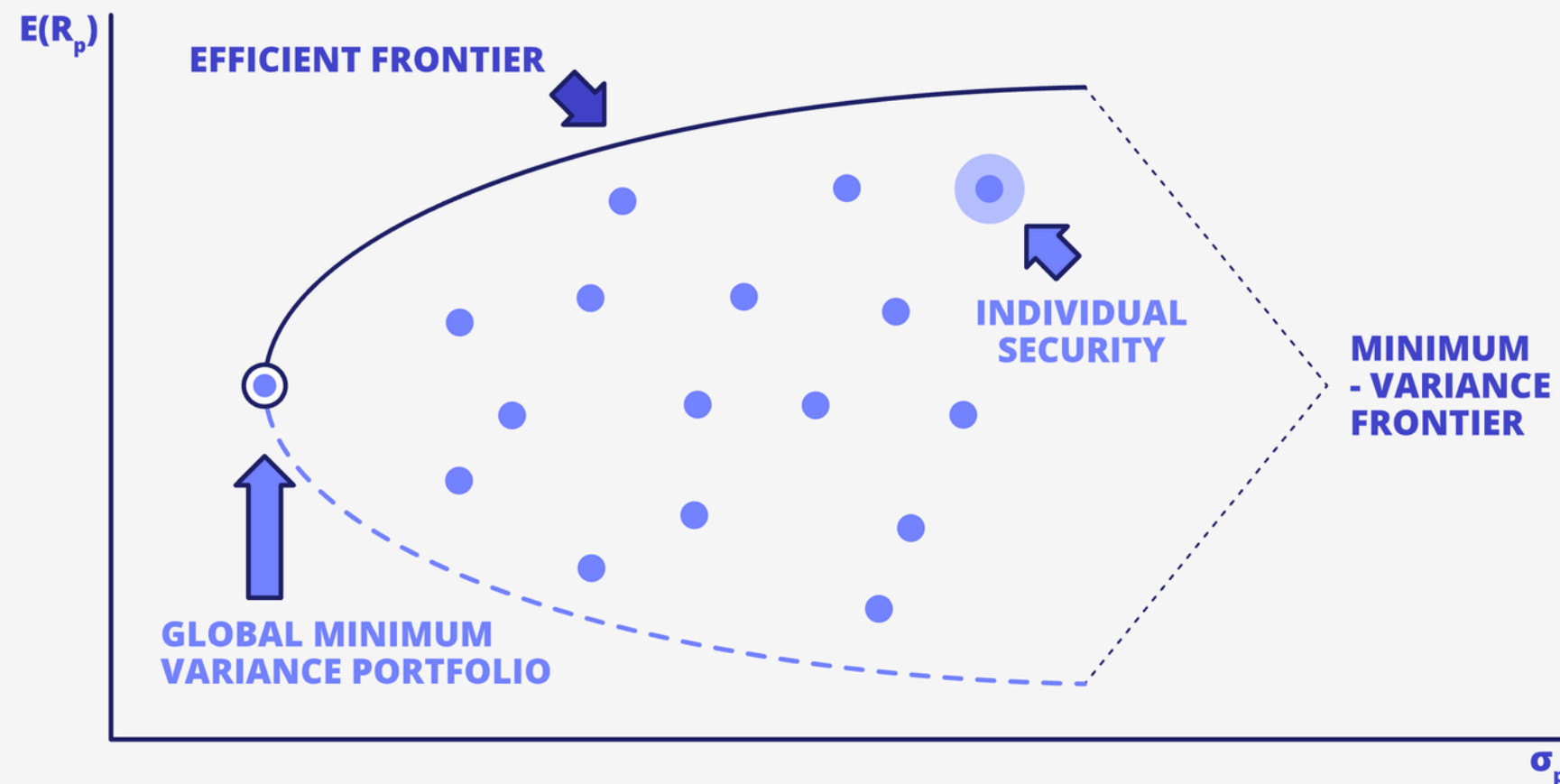
R_f = Risk-Free rate

σ_p = Standard deviation of portfolio's excess return

Min Volatility

A **minimum volatility portfolio (MVP)** that prioritizes **absolute risk reduction**; this strategy constructs the portfolio with the lowest possible overall price fluctuations by allocating capital to stable, uncorrelated assets, entirely ignoring expected future returns. Based on Modern Portfolio Theory, this strategy provide **downside protection** during market declines while potentially offering equity-like returns over the long term, making it suitable for **risk-averse investors**.

Global Minimum Variance Portfolio



$$f(x_a, x_b) = \sigma_p^2 = x_a^2 \sigma_a^2 + x_b^2 \sigma_b^2 + 2x_a x_b \sigma_{a,b}$$

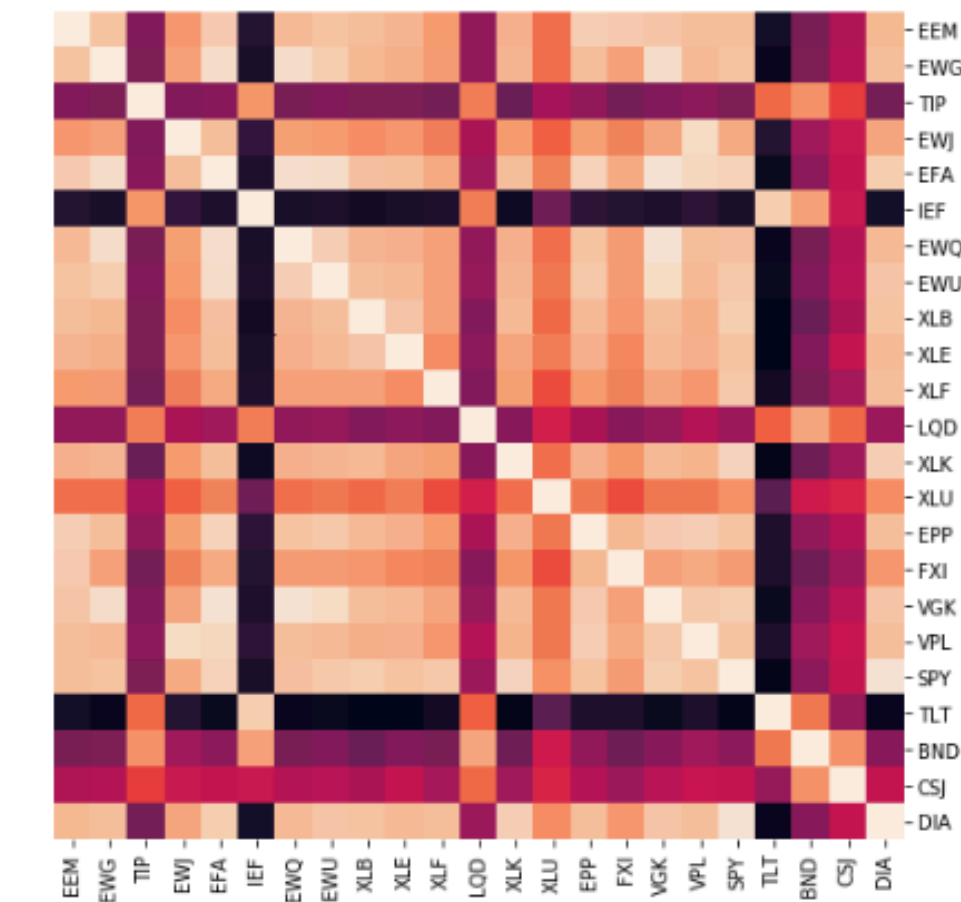
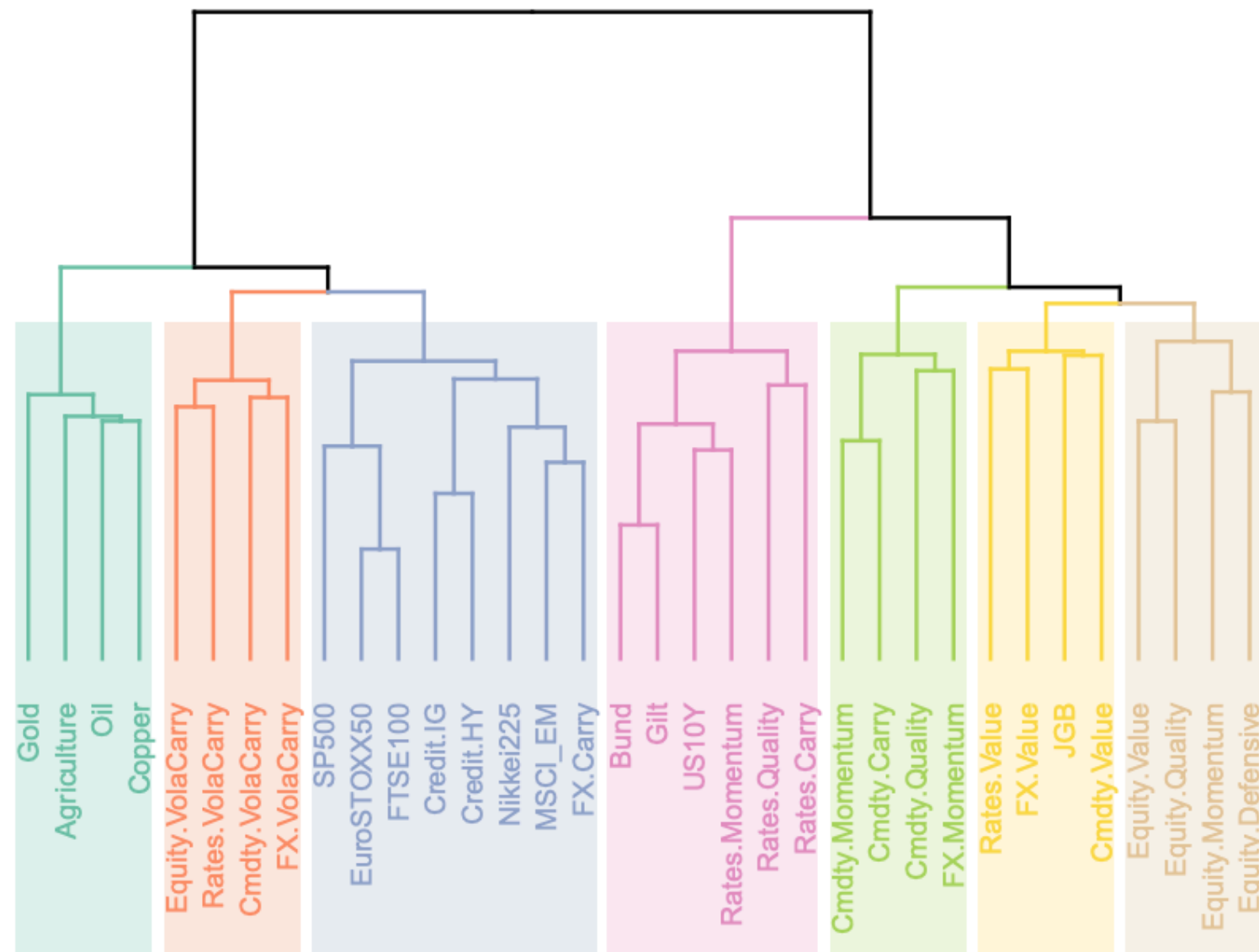
$$x_y = \text{Portfolio Weight of Asset } y$$

$$\sigma_y^2 = \text{Variance of Asset } y$$

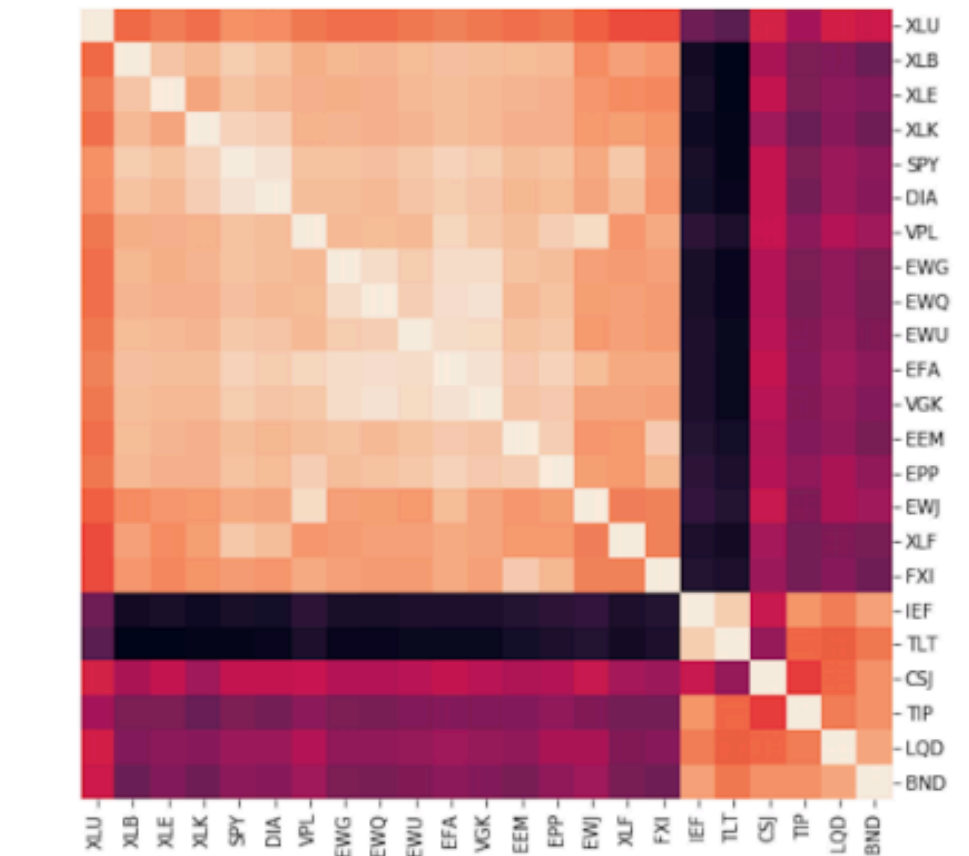
$$\sigma_{y,z} = \text{Covariance Between Asset } y \text{ and Asset } z$$

Hierarchical Risk Parity (HRP)

Hierarchical Risk Parity (HRP), a modern, machine-learning approach to asset allocation; groups assets by their behavioral similarities into correlation clusters, then distributes risk evenly among those groups to create a highly diversified portfolio that avoids the mathematical fragility of traditional optimizers. Does NOT require the **inversion of a covariance matrix**. (less computational complexity).



Unclustered Correlations



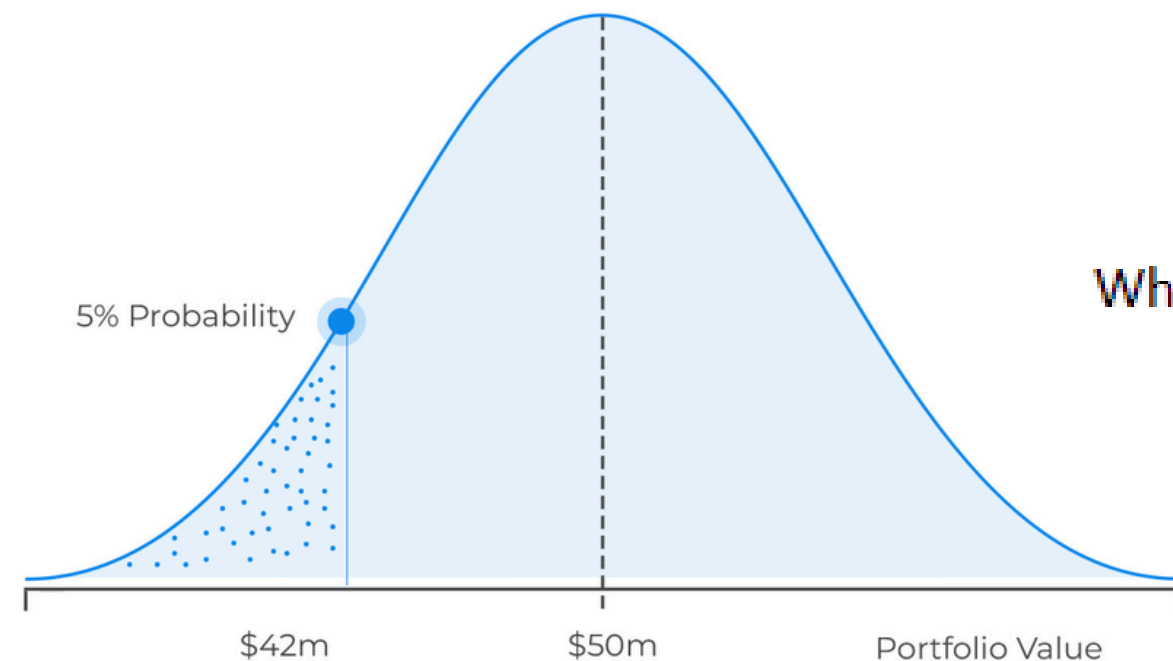
Clustered Correlations

VaR (Value at Risk) - Scaled

A VaR-scaled portfolio uses Value at Risk (VaR) to manage market risk, often employing the "**square root of time**" rule (e.g., scaling 1-day VaR to 10-day VaR) to estimate potential losses over longer horizons. It calculates **maximum expected losses at a given confidence level** (e.g., 95% or 99%) using historical, parametric, or Monte Carlo methods.



5% VaR



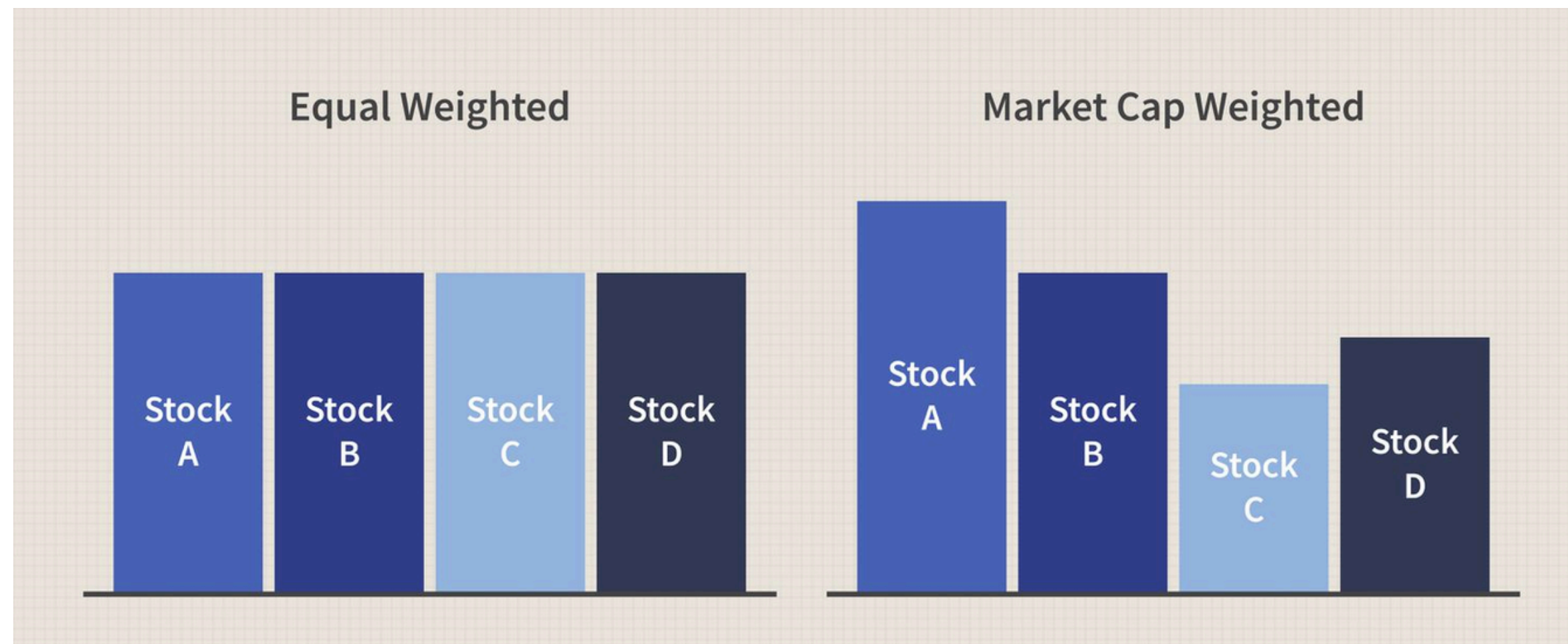
$$\sigma^2 = \frac{1}{n-1} \sum_{t=1}^n (R_t - E(R))^2$$

Where R_t is the rate of return at time t and $E(R)$ is the mean of the return distribution, i.e.

$$E(R) = \frac{1}{n} \sum_{t=1}^n R_t$$

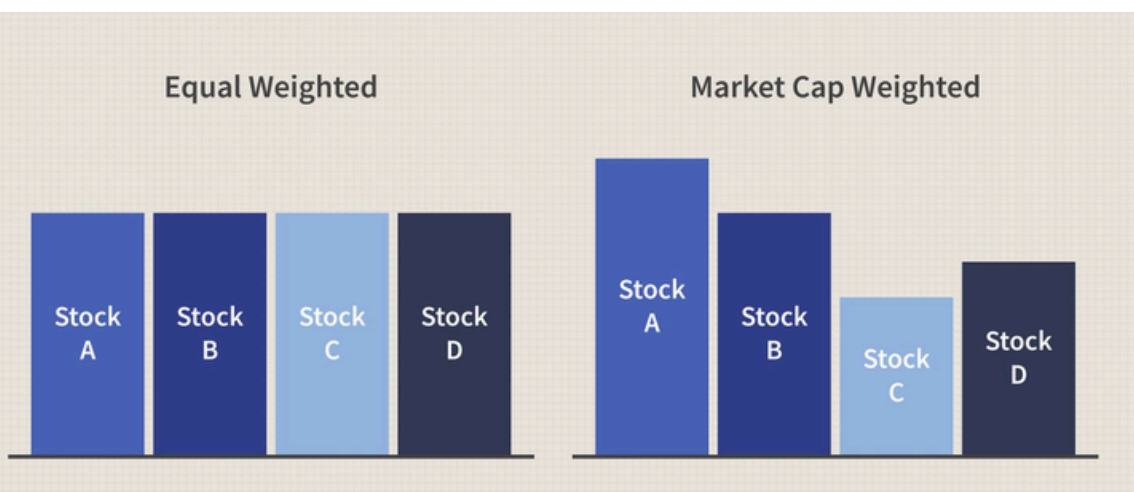
Equal Weight

An equally weighted portfolio assigns the **same percentage of capital to each security**, ensuring small companies have the same impact as large ones. This approach enhances diversification, reduces concentration risk, and historically offers higher returns, though with higher volatility and turnover compared to market-cap-weighted portfolios.



Equal Weight

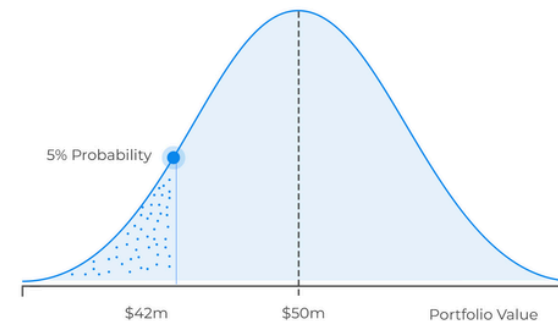
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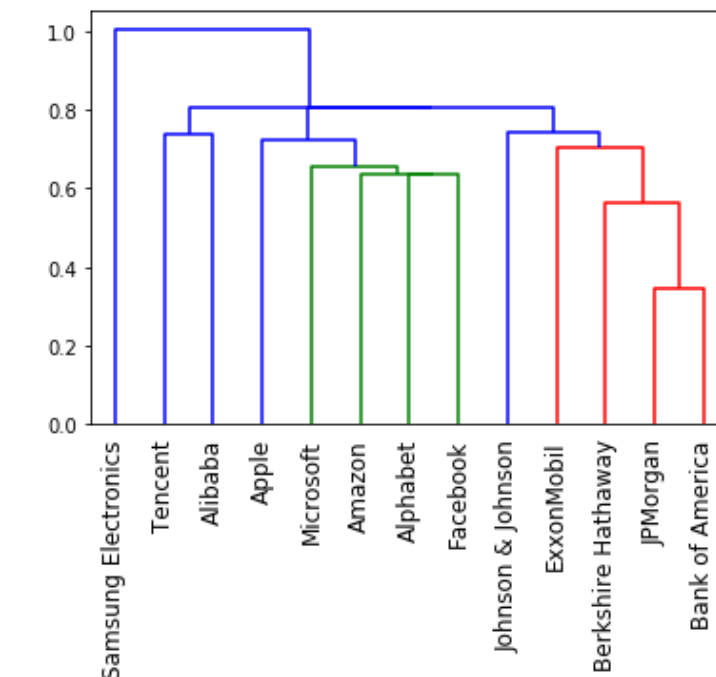
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Hierarchical Risk Parity (HRP)

Hierarchical Risk Parity (HRP) is a modern portfolio optimization technique that uses machine learning (hierarchical clustering) to allocate assets based on risk parity without requiring the inversion of a covariance matrix. It mitigates the instability of traditional mean-variance methods by grouping similar assets into a hierarchical tree structure, resulting in highly diversified, robust portfolios.



Thank You!

